

		Predicted		
		Terrorism	Non-Terrorism	Neutral
Actual	Terrorism	500	200	300
	Non-Terrorism	150	250	200
	Neutral	50	50	300

Terrorism Class (Support = 500 + 200 + 300 = 1000)

Metric	Formula	Calculation	Result
True Positive Rate (Recall)	$TP / (TP + FN)$	$500 / (500 + 500)$	0.5000
True Negative Rate	$TN / (TN + FP)$	$800 / (800 + 200)$	0.8000
False Positive Rate	$FP / (FP + TN)$	$200 / (200 + 800)$	0.2000
False Negative Rate	$FN / (FN + TP)$	$500 / (500 + 500)$	0.5000
Precision	$TP / (TP + FP)$	$500 / (500 + 200)$	0.7143

Terrorism Class (Actual Terrorism = 1000 documents)

- **TP (True Positive):** Actual Terrorism and Predicted Terrorism = 500
- **FN (False Negative):** Actual Terrorism but Predicted as Non-Terrorism or Neutral = 200 (Predicted Non-Terrorism) + 300 (Predicted Neutral) = 500
- **FP (False Positive):** Actual Non-Terrorism /Neutral but Predicted Terrorism = 150 (Non-Terrorism → Terrorism) + 50 (Neutral → Terrorism) = 200
- **TN (True Negative):** Actual Non-Terrorism /Neutral and Predicted as Non-Terrorism or Neutral =
 $[250 \text{ (Non-Terrorism} \rightarrow \text{Non-Terrorism)} + 200 \text{ (Non-Terrorism} \rightarrow \text{Neutral)}] + [50 \text{ (Neutral} \rightarrow \text{Non-Terrorism)} + 300 \text{ (Neutral} \rightarrow \text{Neutral)}]$
 = 450 + 350 = 800

		Predicted		
		Terrorism	Non-Terrorism	Neutral
Actual	Terrorism	500	200	300
	Non-Terrorism	150	250	200
	Neutral	50	50	300

Non-Terrorism Class (Support = 150 + 250 + 200 = 600)

Metric	Formula	Calculation	Result
True Positive Rate (Recall)	$TP / (TP + FN)$	$250 / (250 + 350)$	0.4167
True Negative Rate	$TN / (TN + FP)$	$1150 / (1150 + 250)$	0.8214
False Positive Rate	$FP / (FP + TN)$	$250 / (250 + 1150)$	0.1786
False Negative Rate	$FN / (FN + TP)$	$350 / (350 + 250)$	0.5833
Precision	$TP / (TP + FP)$	$250 / (250 + 250)$	0.5000

Negative Class (Actual Negative = 600 documents)

- **TP (True Positive)**: Actual Non-Terrorism and Predicted Non-Terrorism = 250
- **FN (False Negative)**: Actual Non-Terrorism but Predicted as Terrorism or Neutral = 150 (Predicted Terrorism) + 200 (Predicted Neutral) = 350
- **FP (False Positive)**: Actual Terrorism /Neutral but Predicted Non-Terrorism = 200 (Terrorism → Non-Terrorism) + 50 (Neutral → Non-Terrorism) = 250
- **TN (True Negative)**: Actual Terrorism /Neutral and Predicted as Terrorism or Neutral =
 $[500 \text{ (Terrorism} \rightarrow \text{Terrorism)} + 300 \text{ (Terrorism} \rightarrow \text{Neutral)}] + [50 \text{ (Neutral} \rightarrow \text{Terrorism)} + 300 \text{ (Neutral} \rightarrow \text{Neutral)}] = 800 + 350 = 1150$

		Predicted		
		Terrorism	Non-Terrorism	Neutral
Actual	Terrorism	500	200	300
	Non-Terrorism	150	250	200
	Neutral	50	50	300

Neutral Class (Support = 50 + 50 + 300 = 400)

Metric	Formula	Calculation	Result
True Positive Rate (Recall)	$TP / (TP + FN)$	$300 / (300 + 100)$	0.7500
True Negative Rate	$TN / (TN + FP)$	$1100 / (1100 + 500)$	0.6875
False Positive Rate	$FP / (FP + TN)$	$500 / (500 + 1100)$	0.3125
False Negative Rate	$FN / (FN + TP)$	$100 / (100 + 300)$	0.2500
Precision	$TP / (TP + FP)$	$300 / (300 + 500)$	0.3750

Neutral Class (Actual Neutral = 400 documents)

- **TP (True Positive):** Actual Neutral and Predicted Neutral = 300
- **FN (False Negative):** Actual Neutral but Predicted as Terrorism or Non-Terrorism = 50 (Predicted Terrorism) + 50 (Predicted Non-Terrorism) = 100
- **FP (False Positive):** Actual Terrorism / Non-Terrorism but Predicted Neutral = 300 (Terrorism → Neutral) + 200 (Non-Terrorism → Neutral) = 500
- **TN (True Negative):** Actual Terrorism / Non-Terrorism and Predicted as Terrorism or Non-Terrorism =
 $[500 \text{ (Terrorism} \rightarrow \text{Terrorism)} + 200 \text{ (Terrorism} \rightarrow \text{Non-Terrorism)}] +$
 $[150 \text{ (Non-Terrorism} \rightarrow \text{Terrorism)} + 250 \text{ (Non-Terrorism} \rightarrow \text{Non-Terrorism)}] = 700 + 400 = 1100$

Classification Report:

Metric	Terror Result	Non- Terror Result	Neutral Result	Final Results
True Positive Rate (Recall)	0.5000	0.4167	0.7500	
True Negative Rate	0.8000	0.8214	0.6875	
False Positive Rate	0.2000	0.1786	0.3125	
False Negative Rate	0.5000	0.5833	0.2500	
Precision	0.7143	0.5000	0.3750	
Support	1000	600	400	2000